
Principal component analysis (PCA)

Launcher: `Toolbox/pca.bat`

Purpose

PCA transforms correlated spectral bands into uncorrelated principal components. Early components usually capture broad brightness and vegetation variation; later components may expose subtle crop or soil contrasts, but may also be dominated by noise. This tool repeats PCA for many different band combinations.

Inputs and parameters

Option	What to provide	Behaviour
Input image	An 8-band WV3 or LEGION GeoTIFF	Band positions are interpreted using the selected sensor. A corrected or pansharpened reflectance product is normally appropriate.
Results folder	Existing folder, or blank	Blank creates PCA in the input image's grandparent directory. A manually selected folder must already exist.
PCA combinations CSV	Optional CSV	Blank generates every 3-, 4-, 5-, 6-, and 7-band combination plus the full 8-band set and writes <code>band_combinations.csv</code> . Supplying a CSV processes only its listed combinations.
Vector layer	Optional clip polygon	Crops each PCA product after calculation.

Option	What to provide	Behaviour
Sensor	WV3 or LEGION	Defines the eight band names and their positions. Default: WV3. See the band table.

Combinations CSV format

The file must have these exact columns:

```
id,bands,positions
1,C-B-G,1-2-3
2,R-RE1-N,5-6-7
```

`id` becomes the output number; `bands` is descriptive; `positions` controls the actual one-based source bands. Separate values with hyphens. An invalid file causes processing to stop. A freshly generated eight-band CSV contains 219 combinations, so a complete run creates 219 potentially large raster products.

Processing behaviour

For every combination, the tool selects the requested bands, standardizes them (centres and scales them), and runs Orfeo ToolBox PCA. It retains as many components as selected input bands. Therefore, a three-band combination produces a three-band PCA raster and the full combination produces eight components. Processing uses 2048 MB as the OTB streaming RAM setting.

Outputs

- `band_combinations.csv` when combinations are generated automatically
- `COMB<id>.tif` for each combination

Outputs are floating-point, DEFLATE-compressed Cloud Optimized GeoTIFFs. Existing `COMB<id>.tif` products are skipped, which allows a long run to resume but does not detect whether settings have changed.

Interpretation

Use `band_combinations.csv` to identify the source bands behind each result. Inspect every component, not just component 1. A component's sign is arbitrary: the same contrast can appear light in one run and dark in another. PCA magnitude is not an archaeological probability, and components from different combinations are not directly comparable without examining their loadings and histograms. Validate anomalies against original bands, spectral indices, topography, field patterns, and other dates.